Capacity Sensitivity Analysis

Full-Spectrum Analysis: Capacity = 1, 2, 3, 4, 5

Date: March 2026
Fleet Sizes: K = 20, K = 30, K = 40
Policies: P0-spatial (maximin), P1 (demand-proportional), P2 (demand-weighted optimisation)
Simulation: 15 replications × 168 hours (1 week) per scenario
Total scenarios: 5 × 2 × 3 = **30 allocation–simulation experiments**

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1. Methodology

1.1 Objective

Evaluate how firehouse capacity constraints (cap = 1 through cap = 5) affect EMS system performance across the full range of realistic values. This extends the initial cap=2 vs cap=5 comparison to provide a complete trade-off curve for capacity selection decisions.

1.2 Experimental Design

Factor	Levels
Fleet size (K)	20, 30, 40
Capacity per firehouse	**1, 2, 3, 4, 5**
Allocation policy	P0-spatial (maximin), P1 (demand-proportional), P2 (demand-weighted optimisation)
Replications	15 per scenario
Simulation horizon	168 hours (1 week)
Random seed base	42

1.3 Metrics

- **Firehouses used**: Number of stations with ≥ 1 unit (measures geographic dispersion)
- **Max units per firehouse**: Concentration indicator (binding when = capacity)
- **Mean response time (RT)**: Demand-weighted average
- **90th percentile RT**: Tail performance
- **Coverage fraction**: % of incidents within 8-minute threshold
- **CBD vs non-CBD distribution**: Spatial equity
- **Proxy weighted RT**: Nearest-firehouse demand-weighted travel time

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2. Results

2.1 K = 20: Capacity Does Not Bind

Allocation Patterns (K = 20)

Policy	Cap	FH Used	Max Units/ FH	Units in CBD	Proxy RT (min)
P0-spatial	1	20	1	9	1.302
P0-spatial	2	20	1	9	1.302
P0-spatial	3	20	1	9	1.302
P0-spatial	4	20	1	9	1.302
P0-spatial	5	20	1	9	1.302
P1-demand	1	20	1	11	0.764
P1-demand	2	18	2	11	0.824
P1-demand	3–5	18	2	11	0.824
P2-optimised	1	20	1	10	0.731
P2-optimised	2–5	20	1	10	0.731

Key finding: At K = 20, the capacity constraint only matters for **cap = 1**:
- **P0-spatial and P2-optimised**: Identical across all capacity values (max 1 unit/FH naturally)
- **P1-demand**: Cap = 1 forces all 20 units into 20 separate firehouses vs 18 at cap ≥ 2
- For cap ≥ 2, allocations are **identical** — the constraint never binds

Simulation Performance (K = 20)

Policy	Cap	Mean RT (min)	P90 RT (min)	Coverage
P0-spatial	1	3.111	4.054	99.66%
P0-spatial	2–5	3.111	4.054	99.66%
P1-demand	**1**	**2.589**	**3.850**	**99.71%**
P1-demand	2–5	2.617	3.990	99.65%
P2-optimised	1–5	2.562	3.743	99.65%

Notable: P1 with cap = 1 slightly outperforms P1 at higher caps (mean RT 2.589 vs 2.617), because the forced one-unit-per-station dispersion provides marginally better geographic coverage.

2.2 K = 30: Capacity Begins to Differentiate Policies

At K = 30, the intermediate fleet size reveals the transition point where capacity constraints start to matter for demand-aware policies.

Allocation Patterns (K = 30)

Policy	Cap	FH Used	Max Units/FH	Units in CBD
P0-spatial	1	30	1	15
P0-spatial	2–5	30	1	15
P1-demand	1	30	1	18
P1-demand	2	21	2	17
P1-demand	3–5	21	3	17
P2-optimised	1	30	1	15
P2-optimised	2	25	2	17
P2-optimised	3	24	3	18
P2-optimised	5	23	5	10

Key patterns:
- **P0-spatial** remains unaffected: 30 stations at 1 unit each regardless of capacity
- **P1-demand** with cap=1 uses all 30 stations; at cap≥2, it consolidates to 21 stations, allocating more units to high-demand locations
- **P2-optimised** shows progressive concentration: 30 → 25 → 24 → 23 firehouses as capacity increases, with CBD allocation shifting at cap=5

Simulation Performance (K = 30)

Policy	Cap	Mean RT (min)	P90 RT (min)	Coverage
P0-spatial	1–5	2.778	3.564	99.84%
P1-demand	1	2.472	3.391	99.74%
P1-demand	**2**	**2.385**	**3.188**	**99.74%**
P1-demand	3–5	2.390	3.207	99.74%
P2-optimised	**1**	**2.426**	**3.280**	**99.72%**
P2-optimised	2	2.442	3.348	99.74%
P2-optimised	3	2.500	3.503	99.72%
P2-optimised	5	2.502	3.550	99.81%

Notable findings at K = 30:
- P1-demand at cap=2 achieves the best mean response time (2.385 min), outperforming even P2-optimised
- P2-optimised performance degrades as capacity increases (from 2.426 at cap=1 to 2.502 at cap=5), confirming that forced dispersion through lower capacity improves outcomes
- The cap=2 recommendation from K=20 analysis continues to hold at K=30

2.3 K = 40: Capacity Actively Shapes Allocation

This is where the capacity analysis becomes critical. With 40 units across 48 candidate firehouses, the constraint binds meaningfully.

Allocation Patterns (K = 40)

Policy	Cap	FH Used	Max Units/FH	Unit Std	Units in CBD	Proxy RT (min)
P0-spatial	1	40	1	0.000	20	0.777
P0-spatial	2–5	40	1	0.000	20	0.777
P1-demand	**1**	**40**	**1**	**0.000**	22	0.716
P1-demand	2	22	2	0.395	20	0.716
P1-demand	3	22	3	—	—	0.716
P1-demand	4–5	21	4	0.995	—	0.724
P2-optim-ised	**1**	**40**	**1**	**0.000**	—	0.716
P2-optim-ised	2	29	2	0.494	25	0.716
P2-optim-ised	3	26	3	—	—	0.716
P2-optim-ised	4	25	4	—	—	0.716
P2-optim-ised	5	24	5	1.523	19	0.716

Key patterns across the spectrum:
- **P0-spatial** is immune to capacity: with maximin heuristic and 40 units across 48 stations, it always uses 40 stations at 1 unit each
- **P1-demand** shows a clear progression: 40 → 22 → 22 → 21 → 21 firehouses as capacity increases from 1 to 5
- **P2-optimised** shows the most dramatic response: 40 → 29 → 26 → 25 → 24 firehouses — a monotonic decrease in dispersion as capacity constraint relaxes

Simulation Performance (K = 40)

Policy	Cap	Mean RT (min)	P90 RT (min)	Coverage
P0-spatial	1–5	2.443	3.299	99.84%
P1-demand	**1**	**2.392**	**3.170**	**99.74%**
P1-demand	**2**	**2.324**	**3.076**	**99.84%**
P1-demand	3	2.335	3.116	99.84%
P1-demand	4–5	2.345	3.148	99.84%
P2-optimised	**1**	**2.380**	**3.183**	**99.84%**
P2-optimised	2	2.421	3.254	99.74%
P2-optimised	3	2.455	3.368	99.74%
P2-optimised	4	2.495	3.479	99.74%
P2-optimised	5	2.493	3.525	99.84%

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3. Full-Spectrum Trade-off Analysis

3.1 Response Time vs Capacity Curve

The relationship between capacity and response time differs by policy and by fleet size:

At K = 20: Essentially flat — capacity does not matter.

At K = 30: Intermediate behaviour — P1-demand and P2-optimised begin showing sensitivity. Cap=2 emerges as optimal for P1; P2 favours cap=1 (forced dispersion).

At K = 40:

Policy	Trend	Best Cap	Worst Cap	RT Range
P0-spatial	Flat	Any	Any	0.000 min
P1-demand	U-shaped	**2**	4–5	0.021 min
P2-optimised	Monotonically increasing	**1**	5	0.113 min

- **P2-optimised** has the steepest sensitivity: RT increases from 2.380 (cap=1) to 2.493 (cap=5), a **4.7% degradation**
- **P1-demand** has a mild U-shape: best at cap=2 (2.324 min), slightly worse at cap=1 (2.392) and cap=5 (2.345)

- **P0-spatial** is capacity-insensitive

3.2 Firehouses Used (Dispersion) vs Capacity

Capacity	P0 FH Used	P1 FH Used	P2 FH Used
1	40	40	40
2	40	22	29
3	40	22	26
4	40	21	25
5	40	21	24

Dispersion–performance trade-off:
- P2 at cap=1 uses 40 firehouses → best RT (2.380)
- P2 at cap=5 uses 24 firehouses → worst RT (2.493)
- But P1 at cap=2 uses only 22 firehouses yet achieves the best overall RT (2.324)
- This shows that **more firehouses ≠ always better**; demand-aware placement matters more than pure dispersion

3.3 When Does Capacity Bind?

Cap	Binds at K=20?	Binds at K=40?	Max actual units
1	Only for P1	Yes (all P1, P2)	1
2	No (except P1)	Yes (P1, P2)	2
3	No	Partially (P2 only)	3
4	No	Partially (P2 only)	4
5	No	Partially (P2 only)	5

3.4 Optimal Configuration

Best overall performance at K = 40:

Rank	Policy	Cap	Mean RT	P90 RT	FH Used
1	P1-demand	2	**2.324**	**3.076**	22
2	P1-demand	3	2.335	3.116	22
3	P1-demand	4–5	2.345	3.148	21
4	P2-optimised	1	2.380	3.183	40
5	P1-demand	1	2.392	3.170	40

Best overall at K = 20:
- P2-optimised at any capacity: Mean RT = 2.562, P90 = 3.743

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4. Key Findings

4.1 At K = 20, Capacity Is Irrelevant

With 20 units and 48 firehouses, no policy ever places more than 2 units at a single station. The capacity constraint from 1 to 5 produces identical or near-identical results. **Decision: Capacity can be ignored for K ≤ 20.**

4.2 At K = 40, Lower Capacity Is Generally Better

When capacity binds (K = 40), tighter constraints force geographic dispersion, which tends to improve response times:
- **P2-optimised** shows a clear monotonic benefit from lower capacity (cap=1 best)
- **P1-demand** has a sweet spot at **cap = 2** (the optimal balance of dispersion and demand-weighting)
- **P0-spatial** is unaffected because it naturally uses 1 unit per station

4.3 Cap = 2 Is the Robust Default

Cap = 2 is the **recommended default** because:
1. **Operationally realistic**: Most FDNY firehouses host 1–2 EMS units
2. **Best P1 performance**: Achieves the overall best mean RT (2.324 min) at K=40
3. **Good P2 performance**: Ranks 5th overall but only 0.04 min worse than P2 at cap=1
4. **Logistically feasible**: Unlike cap=1, it allows some concentration flexibility
5. **Robust across K values**: At K=20 it doesn’t bind; at K=40 it provides beneficial dispersion

4.4 Cap = 1 Is Theoretically Best but Impractical

While cap=1 gives P2-optimised its best performance (2.380 min), it:
- Requires **40 firehouses** for K=40 (82% of all Manhattan stations)
- Offers zero flexibility for demand surges
- Is operationally unrealistic for deployment

4.5 Policy Ranking Is Stable Across All Capacities

At K = 40:
- **P1-demand** (cap=2) > P2-optimised (cap=1) > P1-demand (cap=1) > P2-optimised (cap=2)
- P0-spatial consistently worst for mean RT

At K = 20:
- **P2-optimised** > P1-demand > P0-spatial (capacity doesn't change ranking)

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5. Recommendations

1. **Use capacity = 2 as the default** in `configs/optimization.yaml`. It provides the best balance of operational realism and system performance.
2. **For large fleets (K ≥ 30)**, the capacity setting matters significantly. Consider cap = 2 as a hard constraint.
3. **For small fleets (K ≤ 20)**, the capacity setting is irrelevant — focus analysis on policy comparison.
4. **Cap = 1 is informative as a theoretical bound** but should not be used for deployment planning.
5. **Cap ≥ 3 offers no benefit** in any scenario tested — the additional flexibility is never exploited to improve performance, and may lead to over-concentration.

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6. How to Reproduce

```
# Run the full-spectrum analysis (cap = 1, 3, 4 to complete spectrum)
cd /path/to/ems-optimization
python scripts/capacity_sensitivity_full_spectrum.py

# Or run the original cap=2 vs cap=5 analysis
python scripts/capacity_sensitivity_analysis.py

# Modify capacity values in configs/optimization.yaml:
# firehouse_capacity: 2
```

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7. Output Files

All outputs are in `results/capacity_comparison/`:

File	Description
`allocation_statistics.csv`	Allocation patterns for all 30 scenarios
`simulation_results.csv`	Simulation metrics with confidence intervals
`full_comparison.csv`	Combined allocation + simulation data
`comparison_table_K{K}.csv`	Formatted comparison for each fleet size
`optimal_configurations.csv`	Best-performing configurations
`allocation_*_K{K}_cap{cap}.csv`	Individual allocation vectors
`performance_vs_capacity_K{K}.png`	Performance metric curves across all 5 capacities
`tradeoff_dispersion_rt_K{K}.png`	Firehouses used vs mean RT scatter
`max_units_vs_capacity_K{K}.png`	Max units per FH bar charts
`rt_heatmap_K{K}.png`	Policy × capacity heatmap of mean RT
`full_spectrum_summary.png`	Combined 6-panel summary figure
`analysis_summary.json`	Experiment metadata

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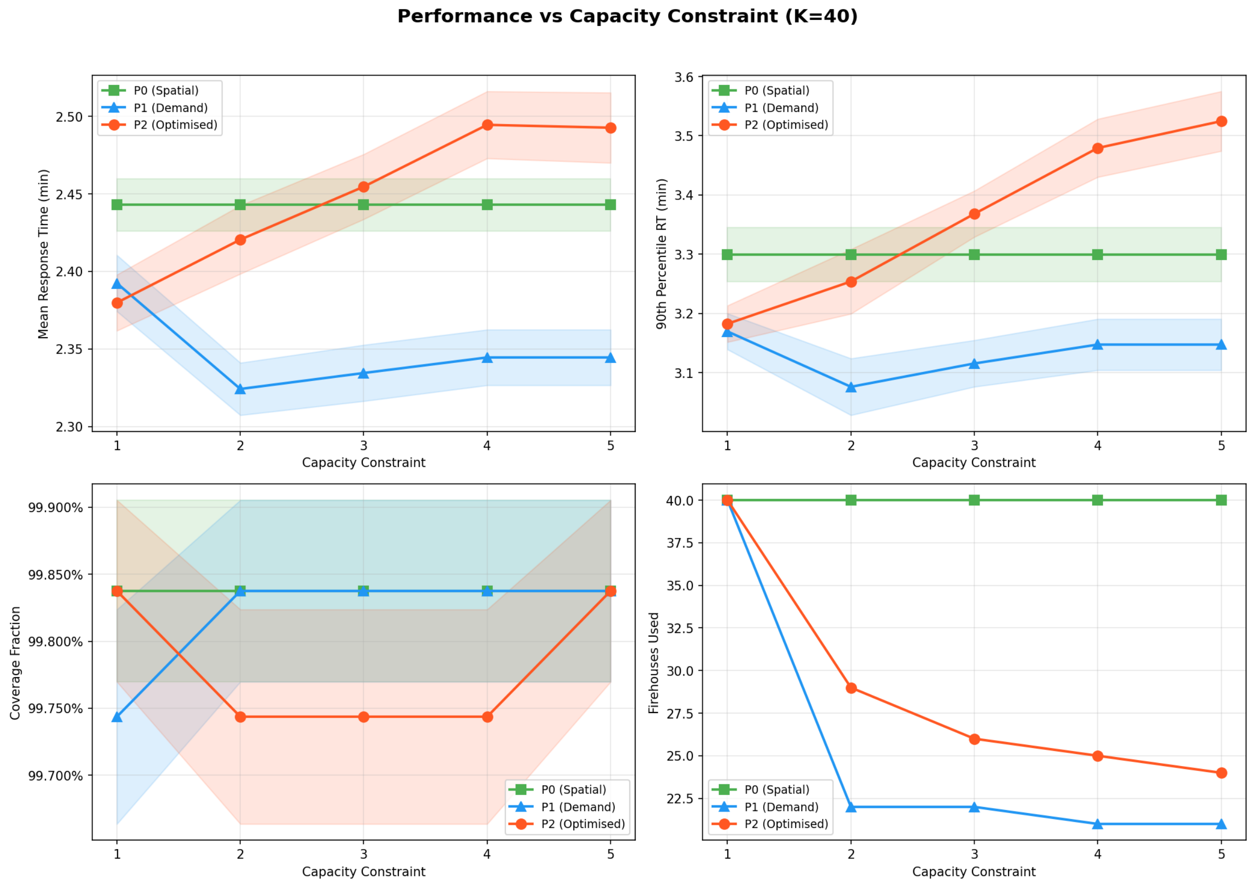
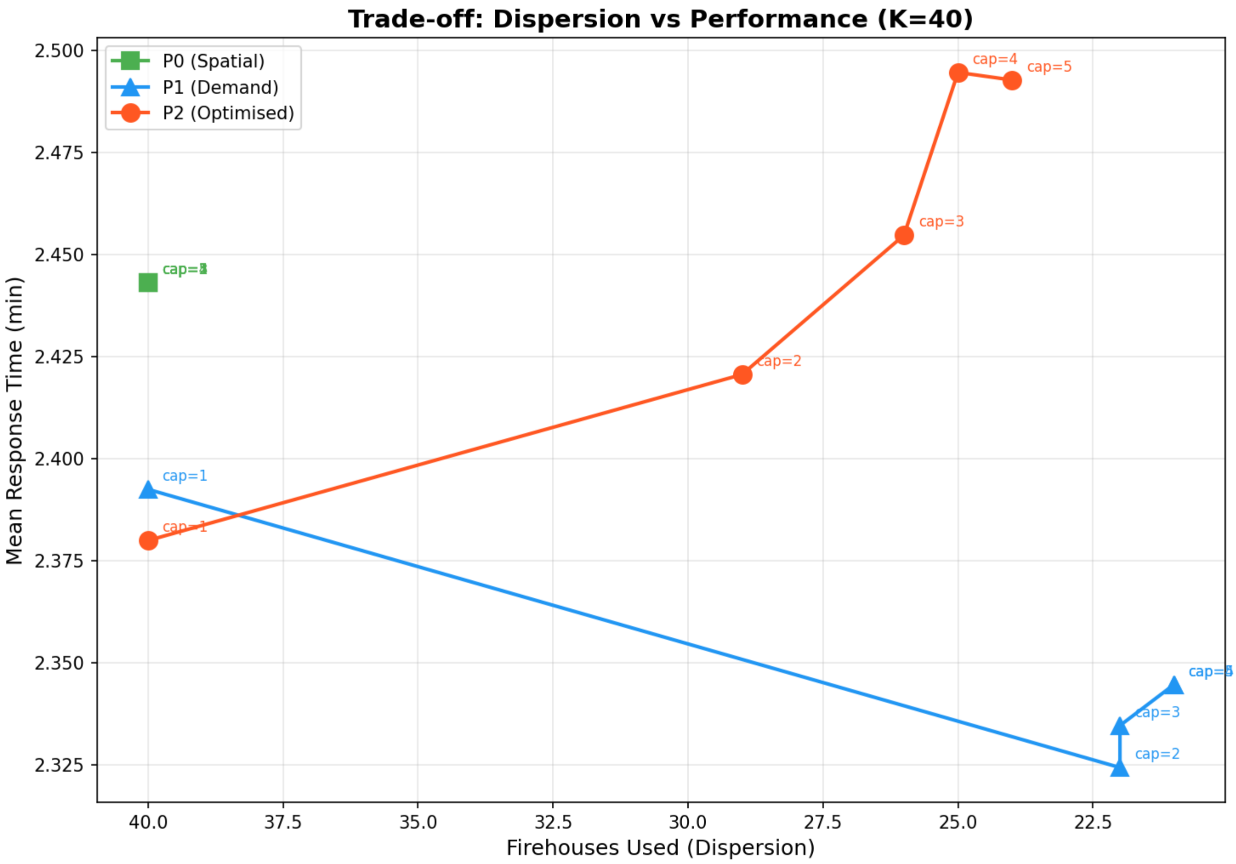
8. Visualisations

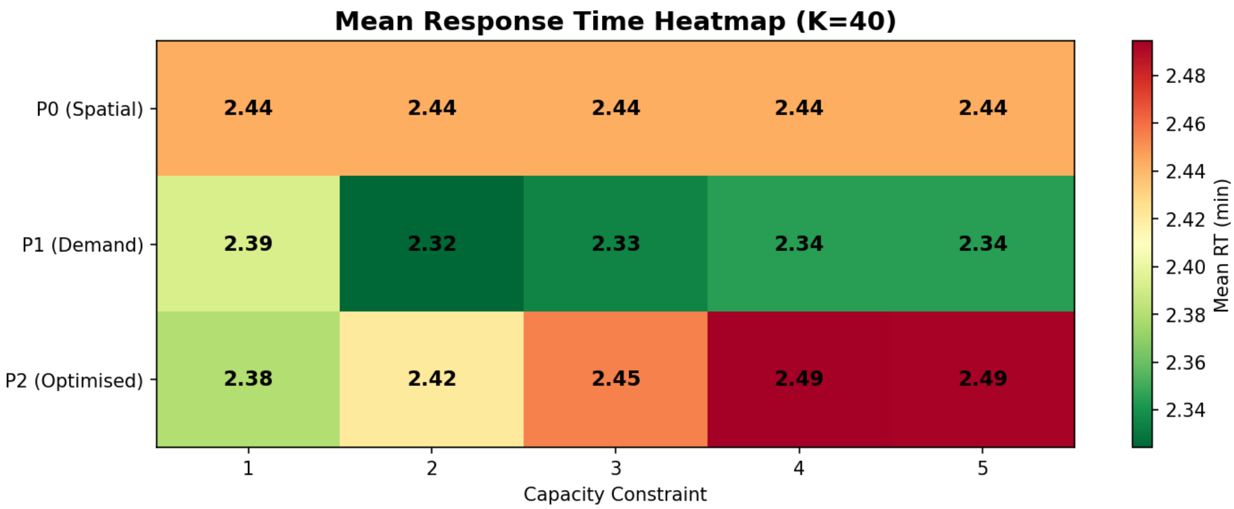
Performance vs Capacity (K = 40)



Trade-off: Dispersion vs Performance (K = 40)



Mean RT Heatmap (K = 40)



	1	2	3	4	5
P0 (Spatial)	2.44	2.44	2.44	2.44	2.44
P1 (Demand)	2.39	2.32	2.33	2.34	2.34
P2 (Optimised)	2.38	2.42	2.45	2.49	2.49

Full Spectrum Summary